Practice workbook · Reading the maximum or minimum from completed-square form

Reading the maximum or minimum value of a quadratic, and the x-value where it happens, from its completed-square form, including simple related functions.

21 question parts: 1 worked-example retry and 20 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-QUA-04

1. Question 7 · part b.ii · 2 marks

2020 NOV · Paper 01

Worked-example retry · Attempt part b.ii only.

7

$$f(x) = x^2 - 9x + 14$$

Given that $f(x)$ can be written in the form $(x + a)^2 + b$, where a and b are constants,

(a) find the value of a and the value of b.

(2)

(b) Hence, or otherwise, find

(i) the minimum value of $f(x)$

(ii) the value of x for which this minimum occurs.

(2)

The curve C has equation $y = f(x)$

The line l has equation $y = x + 5$

(c) Use algebra to find the coordinates of the points of intersection of C and l.

(4)

(d) Use algebraic integration to find the exact area of the finite region bounded by C and l.

(5)

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1. Question 7 · part b.ii · 2 marks · Working and answer

Working and answer

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2. Question 10 · part b.i · 0 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part b.i only.

10 $f(x) = 6x - x^2 \qquad x \in \mathbb{R}$

Given that $f(x)$ can be written in the form $D(x + E)^2 + F$ where D, E and F are integers,

(a) find the value of D, the value of E and the value of F.

(3)

(b) Find

(i) the maximum value of $f(x)$,

(ii) the value of x for which the maximum occurs.

(2)

The curve C has equation $y = f(x)$

The curve S has equation $y = x^2 - 4x + 8$

The curve S intersects the curve C at two points.

(c) Find the coordinates of each of these two points.

(4)

The finite region R is bounded by the curve C and the curve S.

(d) Use algebraic integration to find the area of R.

(4)

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2. Question 10 · part b.i · 0 marks · Working and answer

Working and answer

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3. Question 10 · part b.ii · 2 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part b.ii only.

10 $f(x) = 6x - x^2 \qquad x \in \mathbb{R}$

Given that $f(x)$ can be written in the form $D(x + E)^2 + F$ where D, E and F are integers,

(a) find the value of D, the value of E and the value of F.

(3)

(b) Find

(i) the maximum value of $f(x)$,

(ii) the value of x for which the maximum occurs.

(2)

The curve C has equation $y = f(x)$

The curve S has equation $y = x^2 - 4x + 8$

The curve S intersects the curve C at two points.

(c) Find the coordinates of each of these two points.

(4)

The finite region R is bounded by the curve C and the curve S.

(d) Use algebraic integration to find the area of R.

(4)

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3. Question 10 · part b.ii · 2 marks · Working and answer

Working and answer

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4. Question 7 · part b.i · 0 marks

2020 NOV · Paper 01

Further practice · Attempt part b.i only.

7

$$f(x) = x^2 - 9x + 14$$

Given that $f(x)$ can be written in the form $(x + a)^2 + b$, where a and b are constants,

(a) find the value of a and the value of b.

(2)

(b) Hence, or otherwise, find

(i) the minimum value of $f(x)$

(ii) the value of x for which this minimum occurs.

(2)

The curve C has equation $y = f(x)$

The line l has equation $y = x + 5$

(c) Use algebra to find the coordinates of the points of intersection of C and l.

(4)

(d) Use algebraic integration to find the exact area of the finite region bounded by C and l.

(5)

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4. Question 7 · part b.i · 0 marks · Working and answer

Working and answer

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5. Question 3 · part b.i · 0 marks

2023 JAN · Paper 02

Further practice · Attempt part b.i only.

3

$$f(x) = 8x^2 + 10x - 3$$

Given that $f(x)$ can be written in the form $A(x + B)^2 + C$ where A, B and C are constants,

(a) find the value of A, the value of B and the value of C.

(3)

(b) Hence, or otherwise, find,

(i) the value of x for which $f(x)$ has a minimum,

(ii) the minimum value of $f(x)$.

(2)

The curve C has equation $y = f(x)$.

(c) Find the x coordinate of each of the points where C crosses the x-axis.

(2)

The straight line l has equation $y = 2x + 13$

(d) Use algebra to find the coordinates of the two points of intersection of C and l.

(4)

Using the same axes and the results of parts (b), (c) and (d),

(e) sketch the curve C and the straight line l.

(2)

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5. Question 3 · part b.i · 0 marks · Working and answer

Working and answer

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6. Question 3 · part b.ii · 2 marks

2023 JAN · Paper 02

Further practice · Attempt part b.ii only.

3

$$f(x) = 8x^2 + 10x - 3$$

Given that $f(x)$ can be written in the form $A(x + B)^2 + C$ where A, B and C are constants,

(a) find the value of A, the value of B and the value of C.

(3)

(b) Hence, or otherwise, find,

(i) the value of x for which $f(x)$ has a minimum,

(ii) the minimum value of $f(x)$.

(2)

The curve C has equation $y = f(x)$.

(c) Find the x coordinate of each of the points where C crosses the x-axis.

(2)

The straight line l has equation $y = 2x + 13$

(d) Use algebra to find the coordinates of the two points of intersection of C and l.

(4)

Using the same axes and the results of parts (b), (c) and (d),

(e) sketch the curve C and the straight line l.

(2)

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6. Question 3 · part b.ii · 2 marks · Working and answer

Working and answer

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7. Question 2 · part b.i · 0 marks

2024 JUN · Paper 01

Further practice · Attempt part b.i only.

2 $f(x) = 2x^2 + 4x + 9$

Given that $f(x)$ can be written in the form $A(x + B)^2 + C$, where A, B and C are integers,

(a) find the value of A, the value of B and the value of C

(3)

(b) Hence, or otherwise, find

(i) the value of x for which $\frac{1}{f(x)}$ is a maximum

(ii) the maximum value of $\frac{1}{f(x)}$

(2)

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7. Question 2 · part b.i · 0 marks · Working and answer

Working and answer

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8. Question 2 · part b.ii · 2 marks

2024 JUN · Paper 01

Further practice · Attempt part b.ii only.

2 $f(x) = 2x^2 + 4x + 9$

Given that $f(x)$ can be written in the form $A(x + B)^2 + C$, where A, B and C are integers,

(a) find the value of A, the value of B and the value of C

(3)

(b) Hence, or otherwise, find

(i) the value of x for which $\frac{1}{f(x)}$ is a maximum

(ii) the maximum value of $\frac{1}{f(x)}$

(2)

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8. Question 2 · part b.ii · 2 marks

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Further practice · Attempt part b.ii only.

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Question 2 continued

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8. Question 2 · part b.ii · 2 marks · Working and answer

Working and answer

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9. Question 2 · part b.i · 0 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part b.i only.

2 $f(x) = 7 + 4x - x^2$

(a) Write $f(x)$ in the form $a - b(x + c)^2$ where a, b and c are integers to be found. **(3)**

(b) Hence, or otherwise, find

(i) the maximum value of $f(x)$

(ii) the value of x for which this maximum occurs. **(2)**

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9. Question 2 · part b.i · 0 marks · Working and answer

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10. Question 2 · part b.ii · 2 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part b.ii only.

2 $f(x) = 7 + 4x - x^2$

(a) Write $f(x)$ in the form $a - b(x + c)^2$ where a, b and c are integers to be found. **(3)**

(b) Hence, or otherwise, find

(i) the maximum value of $f(x)$

(ii) the value of x for which this maximum occurs. **(2)**

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10. Question 2 · part b.ii · 2 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part b.ii only.

Question 2 continued

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10. Question 2 · part b.ii · 2 marks · Working and answer

Working and answer

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11. Question 2 · part b.i · 0 marks

2025 NOV · Paper 01

Further practice · Attempt part b.i only.

2 $f(x) = 2x^2 - 10x + 7$

Given that $f(x)$ can be written in the form $f(x) = a(x+b)^2 + c$ where a, b and c are rational numbers,

(a) find the value of a, the value of b and the value of c

(3)

(b) Hence, or otherwise, write down

(i) the minimum value of $f(x)$

(ii) the value of x at which this minimum value occurs.

(2)

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11. Question 2 · part b.i · 0 marks · Working and answer

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12. Question 2 · part b.ii · 2 marks

2025 NOV · Paper 01

Further practice · Attempt part b.ii only.

2 $f(x) = 2x^2 - 10x + 7$

Given that $f(x)$ can be written in the form $f(x) = a(x+b)^2 + c$ where a, b and c are rational numbers,

(a) find the value of a, the value of b and the value of c

(3)

(b) Hence, or otherwise, write down

(i) the minimum value of $f(x)$

(ii) the value of x at which this minimum value occurs.

(2)

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12. Question 2 · part b.ii · 2 marks

2025 NOV · Paper 01

Further practice · Attempt part b.ii only.

Question 2 continued

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12. Question 2 · part b.ii · 2 marks · Working and answer

Working and answer

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13. Question 2 · part b.i · 0 marks

2021 JUN · Paper 02

Further practice · Attempt part b.i only.

2 $f(x) = 2 + \frac{4}{5}x - \frac{1}{25}x^2$

Given that $f(x)$ can be expressed in the form $A - B(x + C)^2$ where A, B and C are constants,

(a) find the value of A, the value of B and the value of C.

(4)

(b) Hence write down

(i) the maximum value of $f(x)$,

(ii) the value of x for which this maximum occurs.

(2)

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13. Question 2 · part b.i · 0 marks · Working and answer

Working and answer

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14. Question 2 · part b.ii · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part b.ii only.

2 $f(x) = 2 + \frac{4}{5}x - \frac{1}{25}x^2$

Given that $f(x)$ can be expressed in the form $A - B(x + C)^2$ where A, B and C are constants,

(a) find the value of A, the value of B and the value of C.

(4)

(b) Hence write down

(i) the maximum value of $f(x)$,

(ii) the value of x for which this maximum occurs.

(2)

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14. Question 2 · part b.ii · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part b.ii only.

Question 2 continued

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14. Question 2 · part b.ii · 2 marks · Working and answer

Working and answer

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15. Question 11 · part b.i · 0 marks

2023 JUN · Paper 01

Further practice · Attempt part b.i only.

11 $f(x) = 10 + 6x - x^2$

Given that $f(x)$ can be written in the form $A(x + B)^2 + C$ where A, B and C are constants,

(a) find the value of A, the value of B and the value of C

(4)

(b) Hence, or otherwise, find

(i) the value of x for which $f(x)$ has its greatest value

(ii) the greatest value of $f(x)$

(2)

The curve C has equation $y = f(x)$

The curve S with equation $y = x^2 - x + 13$ intersects curve C at two points.

(c) Find the x coordinate of each of these two points.

(3)

(d) Use algebraic integration to find the exact area of the finite region bounded by the curve C and the curve S.

(5)

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15. Question 11 · part b.i · 0 marks · Working and answer

Working and answer

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16. Question 11 · part b.ii · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part b.ii only.

11 $f(x) = 10 + 6x - x^2$

Given that $f(x)$ can be written in the form $A(x + B)^2 + C$ where A, B and C are constants,

(a) find the value of A, the value of B and the value of C

(4)

(b) Hence, or otherwise, find

(i) the value of x for which $f(x)$ has its greatest value

(ii) the greatest value of $f(x)$

(2)

The curve C has equation $y = f(x)$

The curve S with equation $y = x^2 - x + 13$ intersects curve C at two points.

(c) Find the x coordinate of each of these two points.

(3)

(d) Use algebraic integration to find the exact area of the finite region bounded by the curve C and the curve S.

(5)

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16. Question 11 · part b.ii · 2 marks · Working and answer

Working and answer

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17. Question 10 · part b · 1 mark

2024 NOV · Paper 01

Further practice · Attempt part b only.

10 $f(x) = 3 - 4x - 9x^2$

Given that $f(x)$ can be expressed in the form $A - B(x + C)^2$ where A, B and C are positive constants

(a) find the value of A, the value of B and the value of C

(4)

(b) Hence write down the maximum value of $f(x)$

(1)

The equation $f(x) = 0$ has roots α and β

Without solving the equation $f(x) = 0$

(c) form a quadratic equation, with integer coefficients, that has roots $\frac{3\alpha}{\beta}$ and $\frac{3\beta}{\alpha}$

(6)

(d) Show that $(x + y)^3 = x^3 + y^3 + 3xy(x + y)$

(1)

$g(x) = 3x^2 + qx + r$ where q and r are constants

The equation $g(x) = 0$ has roots $\alpha^2 - \beta$ and $\beta^2 - \alpha$ where α and β are the roots of the equation $f(x) = 0$

(e) Using your answer to part (d), find in simplified exact form, the value of q and the value of r

(6)

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17. Question 10 · part b · 1 mark · Working and answer

Working and answer

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18. Question 2 · part b.i · 0 marks

2023 NOV · Paper 01

Further practice · Attempt part b.i only.

2 $$g(x) = 2x^2 + \frac{1}{2}x - 3$$

(a) Express g(x) in the form $p(x+q)^2 + r$ where p, q and r are rational numbers to be found.

(3)

(b) Find

(i) the minimum value of g(x)

(ii) the value of x at which this minimum occurs.

(2)

$$h(x) = 2x^6 + \frac{1}{2}x^3 - 3$$

(c) Hence, or otherwise, write down

(i) the minimum value of h(x)

(ii) the value of x at which this minimum occurs.

(2)

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18. Question 2 · part b.i · 0 marks · Working and answer

Working and answer

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19. Question 2 · part b.ii · 2 marks

2023 NOV · Paper 01

Further practice · Attempt part b.ii only.

2 $$g(x) = 2x^2 + \frac{1}{2}x - 3$$

(a) Express g(x) in the form $p(x+q)^2 + r$ where p, q and r are rational numbers to be found.

(3)

(b) Find

(i) the minimum value of g(x)

(ii) the value of x at which this minimum occurs.

(2)

$$h(x) = 2x^6 + \frac{1}{2}x^3 - 3$$

(c) Hence, or otherwise, write down

(i) the minimum value of h(x)

(ii) the value of x at which this minimum occurs.

(2)

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19. Question 2 · part b.ii · 2 marks · Working and answer

Working and answer

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20. Question 2 · part c.i · 0 marks

2023 NOV · Paper 01

Further practice · Attempt part c.i only.

2 $$g(x) = 2x^2 + \frac{1}{2}x - 3$$

(a) Express g(x) in the form $p(x+q)^2 + r$ where p, q and r are rational numbers to be found.

(3)

(b) Find

(i) the minimum value of g(x)

(ii) the value of x at which this minimum occurs.

(2)

$$h(x) = 2x^6 + \frac{1}{2}x^3 - 3$$

(c) Hence, or otherwise, write down

(i) the minimum value of h(x)

(ii) the value of x at which this minimum occurs.

(2)

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20. Question 2 · part c.i · 0 marks · Working and answer

Working and answer

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21. Question 2 · part c.ii · 2 marks

2023 NOV · Paper 01

Further practice · Attempt part c.ii only.

2 $$g(x) = 2x^2 + \frac{1}{2}x - 3$$

(a) Express g(x) in the form $p(x+q)^2 + r$ where p, q and r are rational numbers to be found.

(3)

(b) Find

(i) the minimum value of g(x)

(ii) the value of x at which this minimum occurs.

(2)

$$h(x) = 2x^6 + \frac{1}{2}x^3 - 3$$

(c) Hence, or otherwise, write down

(i) the minimum value of h(x)

(ii) the value of x at which this minimum occurs.

(2)

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21. Question 2 · part c.ii · 2 marks

2023 NOV · Paper 01

Further practice · Attempt part c.ii only.

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Question 2 continued

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(Total for Question 2 is 7 marks)

21. Question 2 · part c.ii · 2 marks · Working and answer

Working and answer

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